# ERIC DECASPER

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## EDUCATION

**California Polytechnic State University | San Luis Obispo, CA** **2022 - Present**
Bachelor of Science in **Mechanical Engineering** Expected Graduation: December 2026

## TECHNICAL SKILLS

- SolidWorks
- Onshape
- Autodesk Fusion 360
- DFM/GD&T
- FEA (ANSYS)
- 3D Printing (FDM/SLA/SLS)
- Python
- Java
- MATLAB

## WORK EXPERIENCE

**Western Magnetics Company** | South San Francisco, CA **June 2026 - August 2026**
*Mechanical Engineering Intern*

- Built the pneumatic vibration-damping fixture and worked on the dashboard for the end-of-line tester that grades every motor for imbalance, vibration, and drag in a 45-second cycle — 4,400+ cycles by August.
- Created an in-line testing process for magnet bond strength that caught an under-cured batch before it shipped, and qualified an adhesive 2–3x stronger, cutting adhesion failures from >70% to near 0%.
- Owned custom wire-EDM tooling from design through fabrication and set up an outside machining partnership that turns around urgent tooling in about a week.
- Traced motor KV variation to magnet thickness and lamination tolerance stack-up, then built a measurement fixture so the team could quantify and control it.

**Voliro AG** | Zurich, Switzerland **July 2025 - December 2025**
*Mechanical Engineering Intern*

- Developed aerial spot-repair payloads with MontiPower from concept through flight-tested prototypes, proving out a find-it/fix-it workflow — inspect and repair on one platform — that Voliro now sells as a service.
- Re-architected legacy payload assemblies in SolidWorks into modular, maintainable CAD structures to support scalable product development and future payload updates and integration.
- Designed and validated a cooling solution for a next-generation drone radio, building a 40°C environmental chamber to verify sustained operation at worst-case ambient temperature.

## ENGINEERING PROJECTS

**Payload R&D, Voliro AG |** Cal Poly SLO **January 2026 - Present**

- Leading an industry-sponsored senior project with Voliro AG: a drone-mounted robotic arm that sweeps an electromagnetic sensor across steel structures to inspect them for defects at height.
- Designed the two-link carbon fiber arm under a 1 kg limit and built its onboard control system — a Teensy microcontroller running inverse kinematics to drive two servos.
- Built and bench-tested the payload on real steel assets; currently integrating it onto a Voliro drone shipped to Cal Poly for independent flight testing.

**Aerial Manipulation Machine Learning Project |** Cal Poly SLO **September 2026 - Present**

- Training imitation-learning controllers (ACT, Diffusion Policy) in Isaac Lab for a drone-mounted arm, benchmarking data requirements toward a learned controller for our senior project payload.

## LEADERSHIP AND INVOLVEMENT

**Homestead High School Robotics, FRC Team 670** | Cupertino, CA **August 2021 - June 2022**
*Mechanical Lead*

- Spearheaded the design and fabrication processes leveraging mills, laser cutters, CNC routers, and 3D printers.
- Designed and integrated a custom vacuum and compressed air system into a CNC router for aluminum cutting.
- Won the FIRST Quality Award and the 2022 Cal Games championship.

## LANGUAGES

- English
- Swiss German (Conversational)
- French (Conversational)